Worksheet 3

Instructions

- Include your full name and Coyote ID on the top right of the first page.
- Submit completed worksheets to Gradescope (no tagging required).
- Submit before 11:59 PM on Wednesday, September 16, 2026

Physics Problem Solving Tips

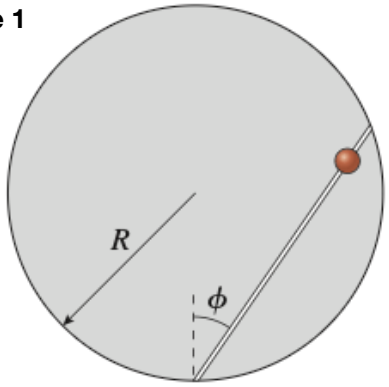
- Draw a diagram (if possible).
- Write down what you know.
- Write down what you want to find.
- First solve symbolically; plug in values at the end.
- Check to make sure your units are correct.
- If you plug in values, check order of magnitude.

Problems

This worksheet covers kinematics in 1D, free fall, and incline planes.

1. A skier is gliding along at $3.0\,\mathrm{m/s}$ on horizontal, frictionless snow. He suddenly starts down a 10° incline. His speed at the bottom is $15\,\mathrm{m/s}$.
   a. What is the length of the incline?
   b. How long does it take him to reach the bottom?
2. A skateboarder starts up a $3.0\,\mathrm{m}$ long ramp at $4.0\,\mathrm{m/s}$. What is the maximum height of the ramp where the skateboarder goes off the end rather than rolling back down?
3. Nicole throws a ball straight up. Chad watches the ball from a window $5.0\,\mathrm{m}$ above the point where Nicole released it. The ball passes Chad on the way up, and it has a speed of $10\,\mathrm{m/s}$ as it passes him on the way back down. How fast did Nicole throw the ball?

Figure 1



4. Figure 1 shows a fixed vertical disk of radius R. A thin frictionless rod is attached to the bottom point of the disk and to a point on the edge, making angle ϕ with the vertical. Find an expression for the time it takes a bead to slide from the top end of the rod to the bottom